Information Visualization

Data vis. on the web

Lesson 7

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Strategies

Your project must include a web page presenting results of your data analysis in an engaging way, showcasing relevant charts. How do we produce web-ready charts?

Jupyter export

Make charts in jupyter, export images or HTML docs

WYSIWYG

Use online tools for data visualizations

Javascript

Use Front-end libraries to plot data directly in web docs

02

Hands-on

Images and charts

What you'll do

Hands-on

Bokeh

Export a chart from a jupyter notebook
Go to the tutorial in GitHub, Colab

RawGraphs

Export a chart from https://www.rawgraphs.io
Go to the tutorial in GitHub, Colab

JS

Create a visualization with Javascript. Go to the tutorial

Exercise

No exercises!

Have a look at the JS libraries you may want to use in your project.

Read more (10 most used js libraries for charting)

03

Data Storytelling

What are the building blocks to create a data-centric story

Questions

Start with a question to frame your story

Context

Audience, objectives

Story

Visual data and presentation techniques

Start with the big Question

Frame the scope

Preliminary question(s) are useful at the beginning when acquiring and filtering data.

The final question is often the result of several attempts, and might not be clear at the beginning.

Example

Examine the relation between {topics} (influence, divergence, correlation)

Explore {topic} over space and time (evolution, potential correlation with other topics/events)

Make data-centric questions

Sub-questions

Decompose the big Question in small, **complementary**, sub-questions.

Answers to such questions, all together, should contribute to answer (or reframe) the big question.

Begin with...

A question must be data-centric, meaning it should start with terms like **where, when, how much, how often.**

N.B. It is hard that visualisations alone can answer questions that start with **why** (which is the domain of data analysis and human interpretation).

Make the right questions

Ask yourself "who"

Your questions should be the ones that your **audience** would do.

Different audiences, different questions.

Ask yourself "why"

Clarify why your audience should be interested in knowing the answer to your questions.

What is their purpose?

What would they do with the information you provide them? How does it *enable* them?

N/N group guidelines
https://www.nngroup.com/articles/persona/
https://www.nngroup.com/articles/journey-mapping-101/

Make answers actionable

SO WHAT?

When you find an answer to any of your sub-questions, ask yourself "so what?"

Example

A pie chart tells you that *"60% of DHDK students like poetry, 30% like novels, and only 10% like plays."*

[audience]
Who cares about this result?

[purpose]
Is this result relevant for any task or future work?

[reframe]
Now that I know this, do I know more about the big Question?

Never ever...

Claim objectivity

Data are abstractions of real-world entities, and **interpretations** are human products, they are not objective.

Always give an account of the **limitations** of your work - it's a plus for credibility, not a weakness.
(some would say "it is a feature, not a bug")

Claim completeness

Rather, look for **representativeness** of data, to justify the validity of your findings.

Reframe (**narrow down**) your scope until you can claim representativeness
(e.g. work on Italian art rather than world-wide art).

Explore or explain

Exploratory

You have some data and **no assumptions**.

You **showcase** the added value of your dataset, which should **reveal** something that was not not known before.

E.g. *"What is the relation between music genres and ethnicity?"*

Explanatory

You have some data and a **thesis** to demonstrate.

You **present** result of your analysis, which **confirms/refutes** your thesis.

E.g. *"Is Jazz Music just Black Music?"*

Explore or explain

Jupyter notebook

It's where you **explore** your data, and show the main features of your dataset.

Then you move to your research questions.

Web project

It's where you **present** your results.

You must provide some information on the dataset that you used for the analysis, but you focus on your questions only here.

Visual data

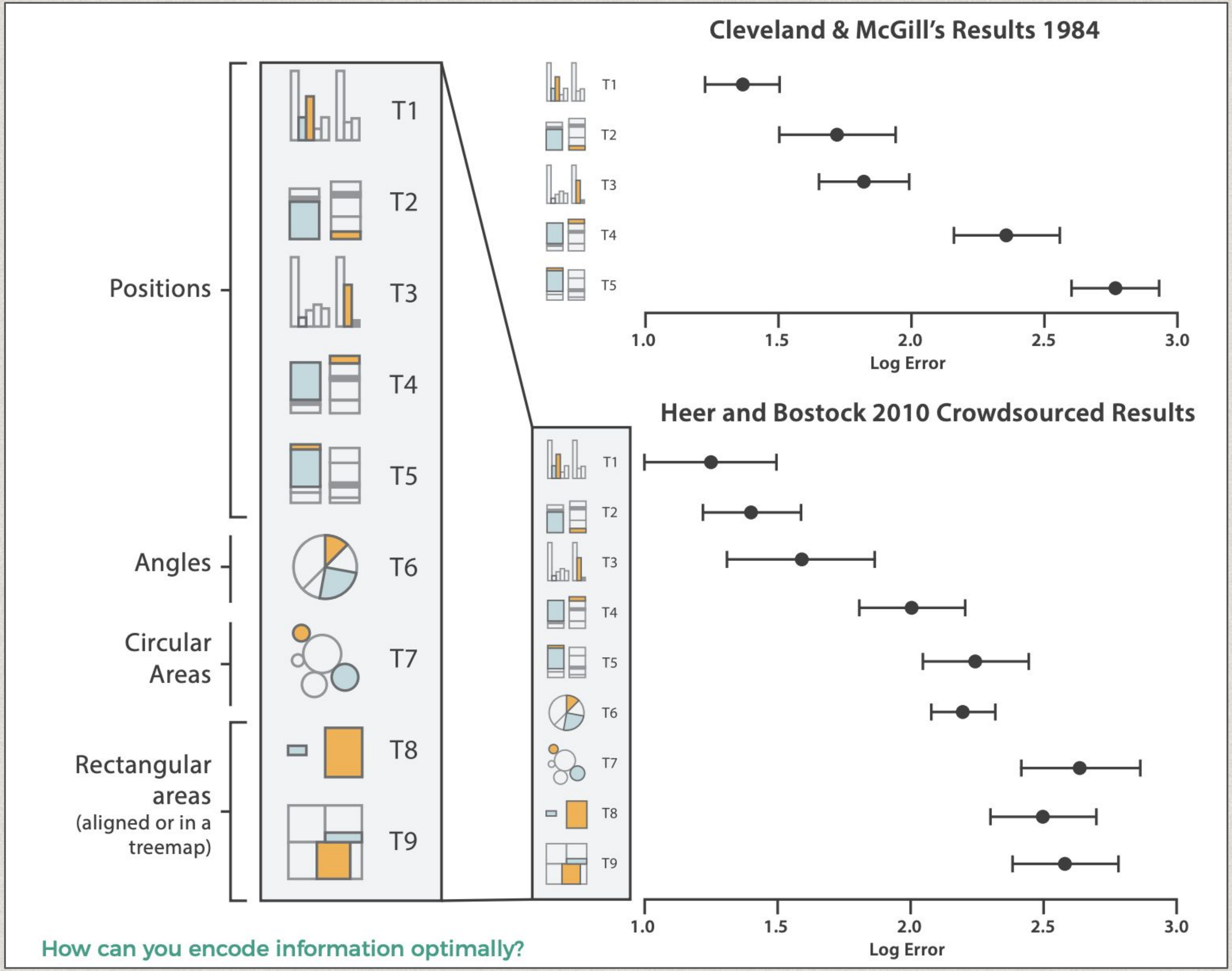
Appropriateness

According to the background of your audience, **some charts** would be easier/harder to understand.

Effectiveness

According to the type of question, the data to be shown and the task, **some charts** would fit better to present a result.

Are these the same charts?

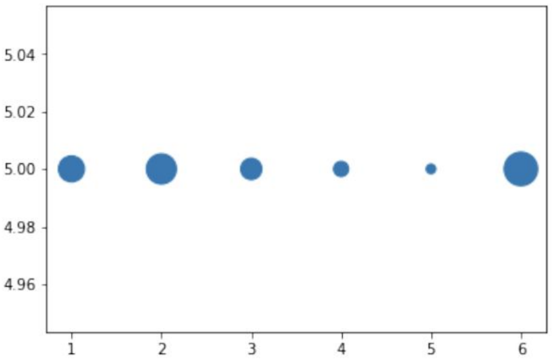


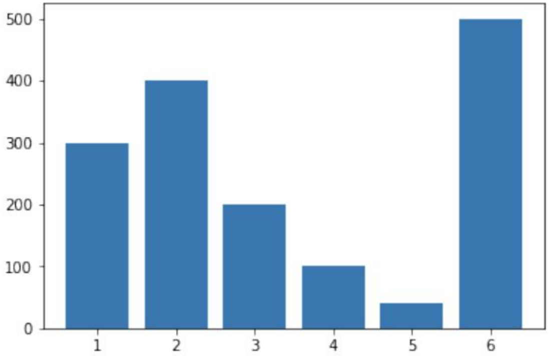
03

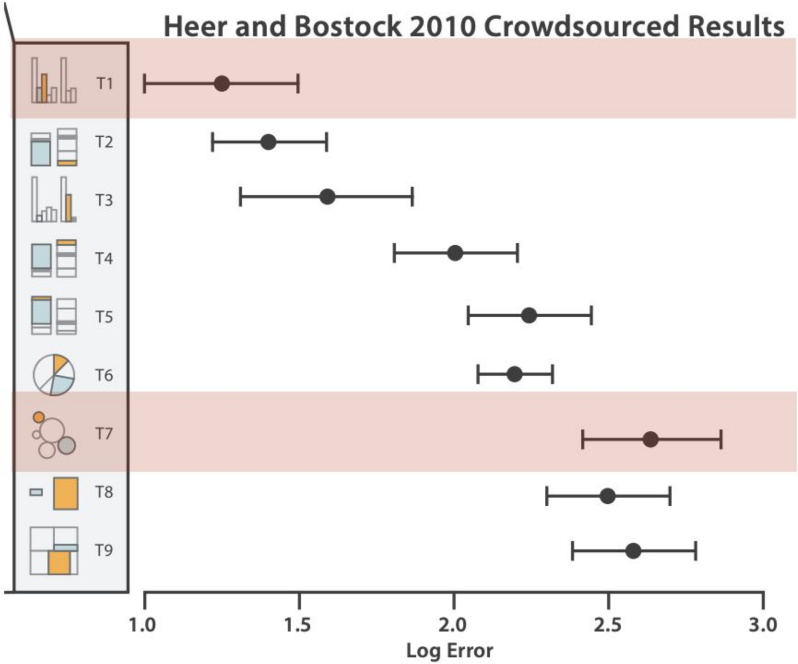
Appropriateness

It is well-known in the literature that the interpretation of some charts is more error-prone than others.

Positions and length (bar charts) are easier for comparison. Differences in areas and angles are more difficult to grasp.







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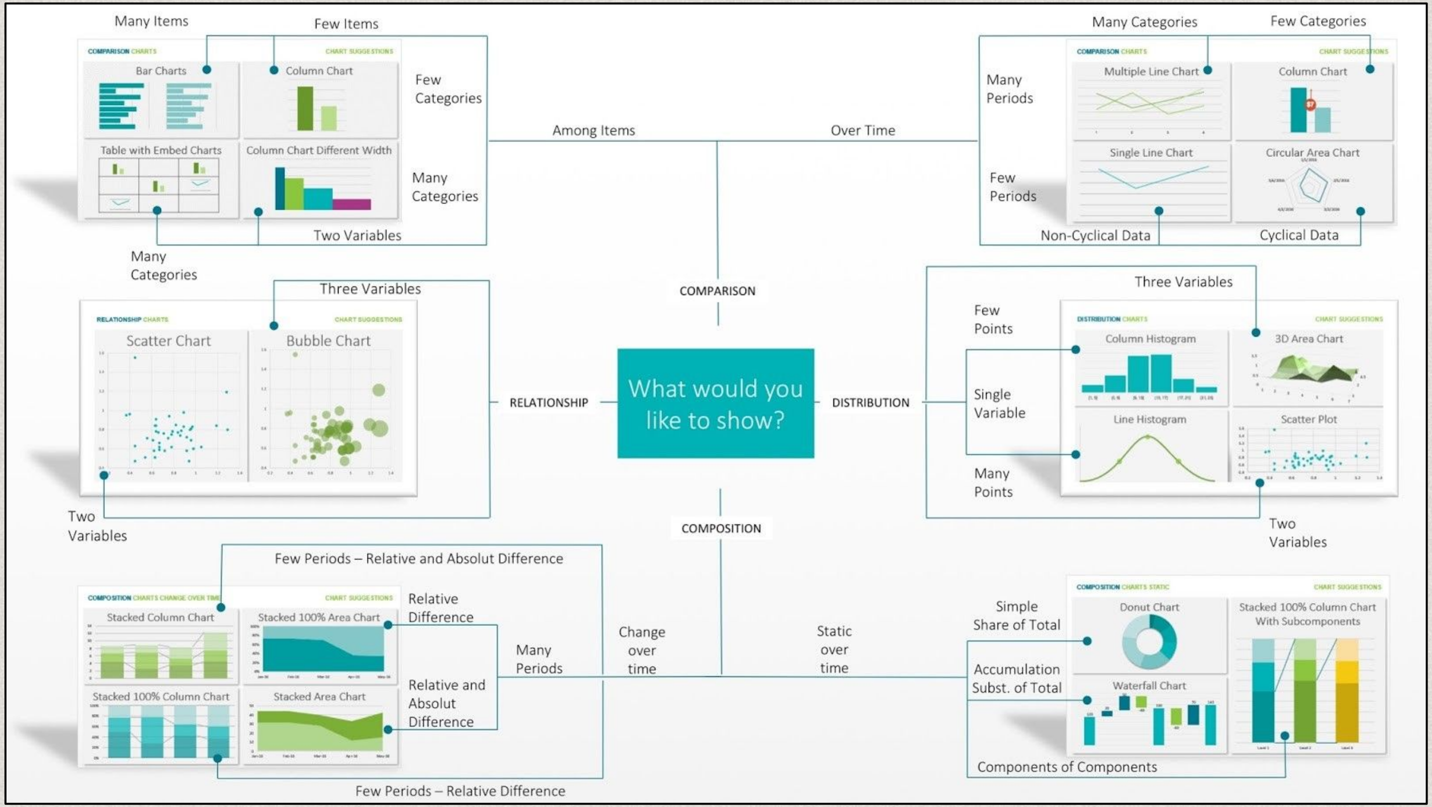
Appropriateness

See an example. Compare the same values in a bubble chart and a bar chart.



03

Effectiveness

Remember
this table?

Choose a chart
according to what
you want your reader
to do

SHOW RELATIONS
COMPARE
SEE DISTRIBUTION
SHOW COMPOSITION

03

Effectiveness

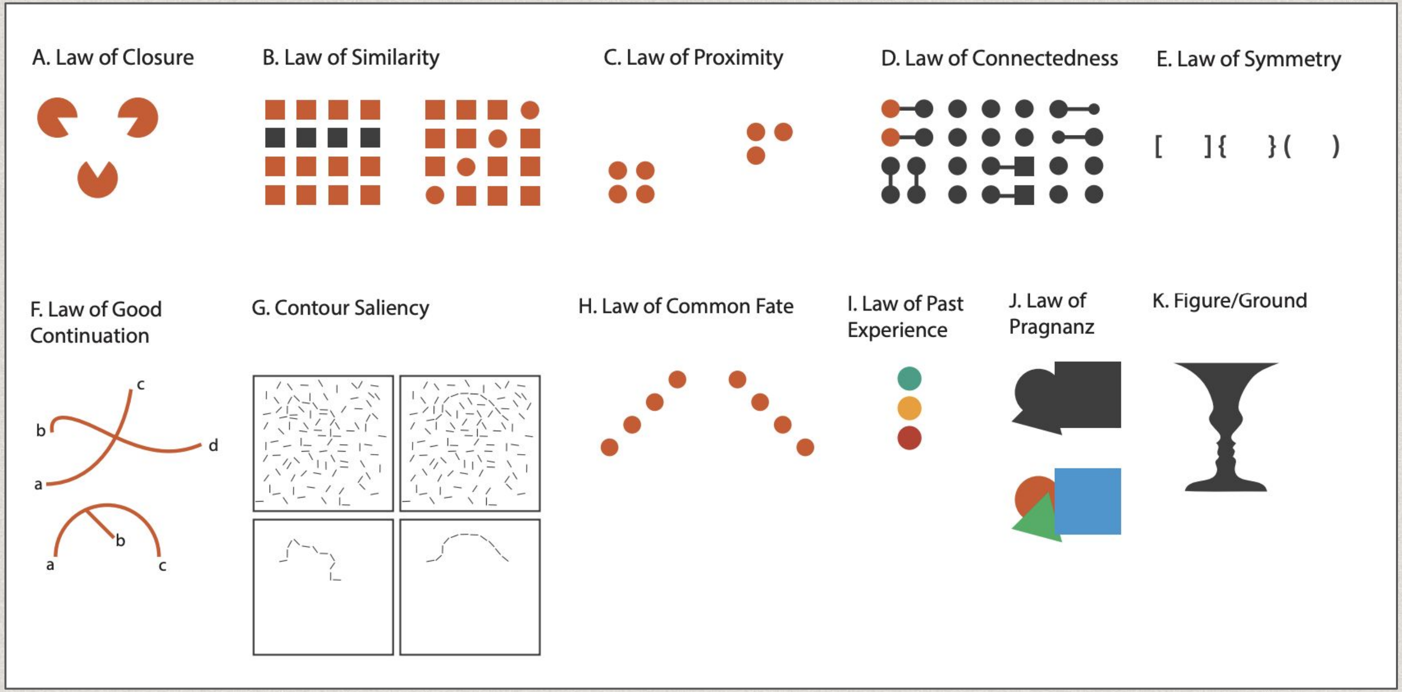
Refine and improve the effectiveness of your presentation using the **Gestalt principles** appropriately.

We look for patterns

We tend to group similar things

We tend to group close things

We tend to group symmetric things



Things arranged in a line or curve are perceived as related

Things that move together are perceived as groups

We use memory to interpret new signals

Light colors pop out, dark colors recede

Small shapes defined by closed contour, texture, color.	Object, idea, entity, node.
Spatially ordered graphical objects.	Related information or a sequence. In a sequence the left-to-right ordering convention is borrowed from written language (English, French, etc.).
Graphical objects in proximity	Similar concepts
Graphical objects having the same shape color, or texture.	Similar concepts
Size, position or height of graphical object	Size, quantity, importance, 2D location
Shapes connected by contour	Related entities, path between entities.
Thickness of connecting contour	Strength of relationship.
Color and texture of connecting contour	Type of relationship.
Shapes enclosed by a contour, a common texture or color	Contained/related entities.
Nested/partitioned regions	Hierarchical concepts.
Attached shapes	Parts of a conceptual structure.

03

Effectiveness

Use a meaningful **semantic mapping** between shapes and patterns that you want to show.

Munzner T. Visualization Analysis & Design. 2014

Choose wisely

Viz. catalogues

- **Data visualisation catalogue** https://datavizcatalogue.com/index.html
- **Visme** https://visme.co/blog/types-of-graphs/
- **Chart maker matrix** https://chartmaker.visualisingdata.com/
- **PolicyViz** https://policyviz.com/books/better-data-visualizations/policyviz-data-visualization-catalog/

Get inspired

Good visualizations examples

- **Information is beautiful awards**
  https://informationisbeautiful.net/
- **Reddit data is beautiful thread**
  https://www.reddit.com/r/dataisbeautiful/
- **Tableau gallery**
  https://public.tableau.com/it-it/gallery/?tab=viz-of-the-day&type=viz-of-the-day

Get inspired

03

Examples

Data storytelling projects

- **Women will**
  https://dataexplorer.womenwill.com/intl/en/thedivide/
- **Where is Poland?**
  https://whereispoland.com/en/where-is-poland/2
- **Lemonade**
  https://www.lemonade.com/giveback-2019
- **Google and NASA**
  https://showcase.withgoogle.com/nasa-fdl/
- **Atlassian - time wasting at work**
  https://www.atlassian.com/time-wasting-at-work-infographic
- **This side of rice**
  http://rice.jennytypes.com/

Thanks!

Do you have any questions?

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https://github.com/marilenadaquino/information_visualization